



The Wrong Turn

Ontario Math C3.2 — read, find, and fix code

Name: _____

Date: _____

★ I can fix a turtle shape loop with the wrong turn angle.

Wrong turn, open shape

In a turtle shape, each shape needs the right turn: square 90, triangle 120, pentagon 72, hexagon 60.

If the turn is wrong, the shape will not close. The range can be right and it still looks wrong — check the turn.

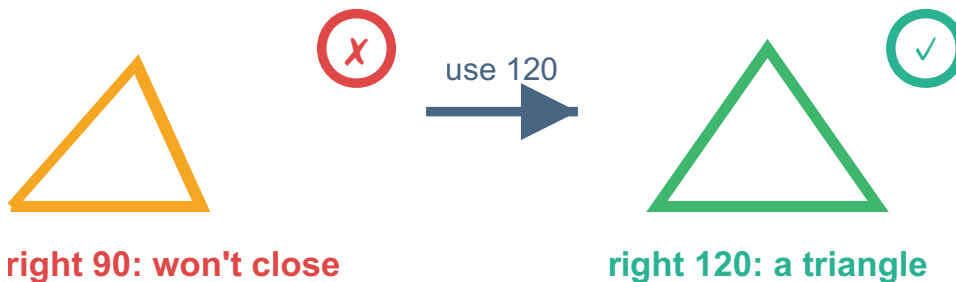
This should draw a TRIANGLE — but it's buggy

```
for i in range(3):  
    bit.forward(50)  
    bit.right(90)
```

PYTHON

OUTPUT

The corners are wrong — the triangle does not close.



right 90 won't make a triangle; right 120 closes it.

1 Find the bug

The triangle code uses right(90). A triangle needs right 120. What should right() be? _____

2 Fix the hexagon

This should draw a hexagon (6 sides): for i in range(6): bit.forward(40); bit.right(90). A hexagon needs right 60. What is wrong, and what should the turn be?

3 Write the fix

Bit should draw a pentagon (5 sides). The turn for a pentagon is 72. Write it: bit.right(_____)

4 Challenge

A SQUARE loop has right(60) instead of right(90). Will the square close? Why not? What should the turn be?

Big idea

Each shape has its own turn. If the shape does not close, the turn is probably wrong — match the turn to the shape (square 90, triangle 120, ...).



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Quick facilitation notes & answer key

What this worksheet teaches

Students debug shape loops where the TURN is wrong, so the shape will not close. The correct turns are given: square 90, triangle 120, pentagon 72, hexagon 60. The method here is SHOULD → DOES → FIX: say what the code SHOULD do, run it to see what it DOES, then FIX the one thing that is wrong. No coding experience needed.

Before you start

No computer needed — the worksheet shows the code and what Bit should do. Optional: a Scratch-style app (for the block loops) or a free Python-turtle playground (for the shapes) to run the code — you only press Run.

How to lead it (about 30 minutes)

1. Show it (you do). Each shape needs its own turn; the wrong turn leaves the shape open.
2. Do one together. Exercise 1 — a triangle needs turn right 120, not 90.
3. Let them work. Students fix a hexagon turn (Ex 2), a pentagon turn (Ex 3), and explain a square that will not close (Ex 4).

Answer key (these are the verified correct answers)

Exercise 1 — turn right 120. Exercise 2 — the turn is wrong; change it to turn right 60.

Exercise 3 — turn right 72. Exercise 4 (challenge) — no, it will not close; turn right 60 is a hexagon turn — a square needs turn right 90.

If students get stuck — and ways to adjust

Watch for: changing the number of sides instead of the turn. Here the sides are right; the turn is wrong.

Make it easier: keep a small chart of the four shape turns nearby.

Make it harder: ask the turn for a shape not on the chart (e.g. eight sides → 45).

Ontario curriculum link (official wording)

C3.1 — solve problems and create computational representations of mathematical situations by writing and executing code, including code that involves sequential, concurrent, and repeating events.

In plain terms: students write and run step-by-step instructions — including a "repeat" instruction — to make something happen.

C3.2 — read and alter existing code, including code that involves sequential, concurrent, and repeating events, and describe how changes to the code affect the outcomes.

In plain terms: students read instructions, find what is wrong, fix it, and explain what the change did.

You'll know it worked when

The child gives each shape its correct turn so it closes.